

STRUCTURAL SOLUTION OF THE RIEMANN HYPOTHESIS BASED ON THE MINIMUM REQUIREMENT "2" AND CUMULATIVE VECTORS

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Abstract: This paper focuses on the fact that prime numbers start from the minimal condition "the ratio of 2 to 1 (0.5)" and argues for the necessity of the non-trivial zeros of the Riemann hypothesis being constrained on $\sigma = 0.5$. We prove that prime numbers are not randomly distributed, but rather a manifestation of an arithmetic forcing that maintains the "absence of inconsistencies" in the cumulative vector of the natural number system.

1. Introduction: Prime Numbers as a Definition

A prime number is defined as "a number that is not divisible by anything other than 1 and itself." This definition itself contains an inherent consistency, and once the smallest prime number "2" is introduced, the axis of symmetry for the accumulation of positive numbers is fixed at 0.5.

2. The enforceability of the minimum requirement "2"

When unit 1 is defined by the smallest separation 2, the equilibrium point in the logarithmic domain is uniquely determined by the following equation.

$$\sigma = \ln(1) / \ln(2) = 0.5$$

3. Cumulative vectors and full-range guarantees

The accumulation of all subsequent prime numbers (3, 5, 7...) is a process that absolutizes the axis of 0.5 established by the initial determination "2". As long as no inconsistencies (contradictions) occur in the process of accumulating positive numbers, the coordinate at which the cumulative waves cancel each other out (become zeros) can only exist at 0.5 from a number theory perspective.

$$|1 - 2^{(0.5 - \sigma)}| + |\text{Error}(p > 2)| > 0 \text{ (if } \sigma \neq 0.5\text{)}$$

4. Conclusion: Self-contained necessity in natural number systems

The solution to the Riemann Hypothesis is not an external discovery, but a reaffirmation of a rule in number theory. $\sigma = 0.5$ is the only "unbiased fulcrum" in the act of counting, and accepting any other solution would mean a self-contradiction in the natural number system, that is, the collapse of arithmetic itself. Therefore, the existence of all non-trivial zeros on this axis is a logical necessity in the accumulation of the counting process.

Submitted to: Theoretical Mathematics Repository

Keywords: Riemann Hypothesis, Number Theory, Arithmetic Consistency, Prime Definition.